

How Often is Politicians' Data Breached?

Evidence from *Have I Been Pwned*

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Abstract

Data breaches involving politicians are concerning because of the threat of impersonation, blackmail, and other nefarious activities. To shed light on the concern, we estimate how frequently politicians' email addresses are compromised. Using a dataset of 12,385 email addresses from politicians across 59 countries spanning three decades, we check whether these addresses are part of breaches using Have I Been Pwned, a widely used online service for searching public breach data. A third of the observed addresses appear in at least one public breach, and over one in five appear in a breach exposing sensitive data such as bank account numbers, biometric data, browsing history, chat logs, or credit card CVV. Decomposing appearances by how the data was exposed shows that most are not compromises of accounts politicians held: 23.5% of addresses appear in broker or scrape aggregation, against 15.9% in a breach of a service the politician plausibly used and 2.32% in corpora derived from infostealer malware. Benchmarked against a general-population sample on a shared catalogue and exposure window, politician addresses appear substantially less often, though differences in breach-corpus coverage leave that comparison unidentified. These are lower bounds: we do not observe every address used by each politician, and HIBP covers only public indexed breaches.

Keywords: Cybersecurity, Have I Been Pwned (HIBP), Elites, Data breaches, Security and privacy

1 Introduction

Data breaches can reveal login credentials, personal details, and service usage patterns, enabling impersonation, extortion, and blackmail. When breaches involve politicians, the stakes are higher ([Harding, 2016](#); [BBC News, 2020](#); [Witman and Mackelprang, 2022](#)). These compromised credentials can give adversaries access to private communications. Such access can be weaponized to influence elections, as in the 2016 DNC hack and the 2017 Macron campaign breach ([Harding, 2016](#); [CrowdStrike, 2016](#); [Almasy, 2017](#); [Shires, 2019](#)). They can also be used for blackmail or to impersonate prominent officials, as in the 2020 Twitter hijackings of Biden and Obama ([BBC News, 2020](#); [Witman and Mackelprang, 2022](#)). Even seemingly innocuous data from old breaches of defunct accounts, like passwords and contact information, enable credential-stuffing attacks on unrelated platforms, which can again expose politicians ([Pal et al., 2022](#); [OWASP Foundation, 2025](#); [Li et al., 2019](#); [Greenberg, 2025](#)). Hence, unlike breaches of ordinary citizens, which primarily harm only the individual, breaches of politicians threaten democratic governance ([Shires, 2019](#)).

Data breaches are also exceedingly common ([Sood and Cor, 2019](#); [Mayer et al., 2021](#); [Sood, 2023](#)). A study of a large random sample of Americans found that more than 80% of people have had their data breached ([Sood and Cor, 2019](#)).¹ While some of the breaches are innocuous, involving already public information, many of the breaches involve private data. For instance, nearly 72% of the public breaches include passwords. Even when only login credentials are compromised, the threat is sizable. Because many people reuse their passwords or only make minor tweaks, attackers using credential tweaking attacks can compromise 83% of the accounts ([Li et al., 2019](#)) (see also [Chintalapati and Sood \(2022\)](#)). The threat is nearly as grave even when leaks involve hashed passwords. Attackers can use resources like [hashes.org](#), which contains clear text values for passwords to crack passwords. In 2021, a study found that nearly 99% of the hashed passwords in the *Have I Been Pwned* (HIBP)

¹A follow-up study using the Florida voter registration data found a similar percentage of people had had their data breached ([Sood, 2023](#)).

database can be recovered (Kanta et al., 2021). These estimates, however, come from the general population rather than elected officials and political elites. They also sit well above the rates we report below, and we take that contrast up directly: [Section 4.4](#) compares our sample against [Sood and Cor \(2019\)](#) on a shared breach catalogue and a shared exposure window, where politician addresses appear in breaches considerably less often than those of US adults. We caution there that uneven national coverage of breach corpora leaves that difference unidentified.

This paper studies the prevalence of politician email-address breaches. We assemble a large dataset of addresses spanning 59 countries over three decades and check them against HIBP. Nearly one in three observed addresses have had at least one breach, and more than one in five have had a “serious” breach, in which private data such as credit-card information may be exposed. These address-level rates are lower bounds rather than population-level estimates for politicians.

Two further results bear directly on how those rates should be read, and we develop both in the Supplementary Information. First, exposure is dominated by third-party aggregation rather than by compromise of politicians’ own accounts: more addresses appear in broker and scrape corpora than in breaches of services the politician plausibly used. Second, on a shared catalogue and a shared exposure window, politician addresses appear in breaches far less often than addresses in a general-population sample. Neither result is a clean estimate of relative risk, because coverage of breach corpora varies by country and address type. Both, however, argue against reading the headline rate as a measure of how often politicians’ accounts are compromised.

2 Research Design, Data, and Measures

Our estimand of interest is the frequency with which politicians’ data, especially sensitive data, is breached. Our population of interest is all politicians who have held a legislative or

executive role at a national, state, or local level in the last three decades. Our definition of population excludes politicians who have never served the government.

To study the question, we assemble a large, diverse convenience sample. Our final data set has thousands of politicians from 59 countries spanning three decades.

To measure how often politicians’ data is leaked, we check public breach data using politicians’ emails. We do this because email addresses allow us to confidently connect a breach to a person. The downside is that our method only enables us to estimate a conservative lower bound of the frequency with which data are leaked for three reasons. First, the breached data must have email addresses. Some public breach data may not; for instance, it may only have usernames and passwords (which can, based on other breaches, be connected back to the politician). Second, we need all the email addresses used by the politician to know all the public data breaches in which a politician’s data has been exposed. We only have official email addresses for many of the politicians. There is also reason to expect that exposure estimated using official email addresses is lower than estimated using personal emails. We expect politicians to be less likely to use official emails when opening online accounts, as the official email accounts for politicians (in many Democracies) are likely temporary. We also expect politicians to be reluctant to associate their official emails with private activities because of disclosure laws and official policies restricting the use of official emails. Lastly, not all breaches become public.

Our data on data breaches comes from HIBP, which collates data from more than 850 public breaches and classifies breached data into nearly 150 categories. Some data breaches are benign, revealing mainly public information; others are more serious, revealing sensitive data. To better understand the actual threat posed to politicians, we classify breaches as serious based on the kind of data breached. Our measure leaves out the most serious breaches. HIBP doesn’t provide data on sensitive breaches—breaches that can be embarrassing, e.g., adult websites like Adult FriendFinder (2015), Ashley Madison, etc.—via its public API. It only allows people who can verify the emails to see those breaches ([Have I Been Pwned](#),

2018).

In all, we start by providing a conservative lower bound of the threat faced by politicians. Next, to account for the attenuating bias stemming from private vs. official emails, we craft a measure distinguishing the two and model the probability of being connected to a data breach. The model yields an adjusted association between address type and exposure rather than a corrected prevalence; we report it as such. We also describe the variation in threat across politicians based on country and gender. Lastly, we analyze how politicians’ data comes to be leaked. We examine which breaches are responsible for the leaks.

2.1 Data on Politicians

Our data on politicians comes from two sources. The first is <https://Everypolitician.org/> (mySociety, 2018). The data has been used extensively by researchers (see, for e.g., Eshima and Smith, 2022; Stockemer and Sundström, 2018; Martini and Walter, 2024; Kosinski et al., 2024) and as a source for building other databases (Kurz and Ettensperger, 2023; Stockemer and Sundström, 2022). The data covers legislatures from across the world between 1997 to 2019.

From the data, we exclude any legislature with fewer than 30 politicians with a listed email address. In all, we have 8,538 unique email addresses from 61 legislatures across 55 countries.

We supplement this data with data gathered by scraping official parliamentary sites of the Argentinian, Brazilian, Danish, Greek, Indian, Nigerian, Norwegian, and Singaporean parliaments (or Senates) and the Indian state legislatures of Bihar, Delhi, Himachal Pradesh, Tamil Nadu, and Uttar Pradesh. We scraped the data in January 2025.

We collate the two datasets and check if the email addresses conform to expected email formats (e.g., RFC 5322-compliant addresses like *name@domain.com*). To check, we first preprocess email strings by stripping whitespace and normalizing character representation by converting them to Unicode (Tauberer, 2024). Next, we check if the domain exists

via DNS resolution and check deliverability by looking up DNS MX records. Emails that fail these checks are removed from the data because querying HIBP with an invalid email (e.g., *an@invalid.email*) will result in a “No breach found” status, biasing our estimates downwards.

Our final dataset has 12,385 unique politician email accounts from 67 legislatures across 59 countries from 1997 to 2025, representing close to 50% of the global population living under electoral regimes (Lührmann et al., 2018; Herre, 2022) and 36% of the global population overall. Table 1 summarizes our data.

2.2 Official Vs. Personal Emails

We classified an address as *official* if its domain was government-controlled (e.g., ccTLDs and SLDs such as .gov, .gouv, .gob, admin.ch, nic.in) or contained clear institutional tokens (e.g., “parliament,” “senate,” “congress,” “assembly,” including common non-English variants). We classified an address as *personal* if the domain was a major webmail provider (e.g., Gmail/Yahoo), a non-government commercial domain (e.g., .com, .net, .biz), or a campaign/staff-managed domain lacking government indicators. The full pattern lists are in Tables SI11 to SI14.

Two caveats motivate robustness checks. First, some “official” mailboxes can be role accounts shared by staff; conversely, some personal domains are professionally managed. Second, exposure risk can correlate with the propensity to publish a personal address. We therefore report a role-account sensitivity analysis using transparent local-part prefixes and alternative seriousness thresholds in Table 3. A within-politician comparison would settle this, and it is not available here, though the obstacle is narrower than the absence of any identifier. EveryPolitician assigns an identifier to each of 16,351 person-legislature records and supplies a Wikidata identifier, which resolves to the individual rather than to the seat, for 11,369 distinct people; 68 records share a Wikidata identifier because the same person served in two legislatures. What is missing is a second address. Keyed on the Wikidata identifier,

Table 1. Politician email coverage by country

	ISO	Country	Emails	Legislature years	Chamber	Legislature	Pop. (M)
1	ALB	Albania	140	2009, 2013, 2017	Unicameral	Kuvendi	2.7
2	AND	Andorra	31	2015	Unicameral	Consell General	0.1
3	ARG	Argentina	71	2025	Upper	Parliament	46.9
4	ARM	Armenia	119	2019	Unicameral	National Assembly	2.8
5	AUS	Australia	177	2004–2016	Lower, Upper	House of Representatives, Senate	26.9
6	BEL	Belgium	149	2014	Lower	Chamber of Representatives	11.9
7	BGR	Bulgaria	205	2013, 2014, 2017	Unicameral	National Assembly	6.4
8	BIH	Bosnia and Herzegovina	42	2014	Lower	House of Representatives	3.2
9	BLR	Belarus	59	2016	Unicameral	House of Representatives	9.1
10	BMU	Bermuda	33	2017	Lower	Parliament	—
11	BRA	Brazil	81	2025	Upper	Parliament	217.6
12	BTN	Bhutan	46	2013	Lower	National Assembly	0.8
13	CAN	Canada	432	2011, 2015	Lower, Upper	House of Commons, Senate	40.4
14	CMR	Cameroon	104	2013	Lower	Assemblée Nationale	29.4
15	COL	Colombia	169	2014, 2018	Lower	Cámara de Representantes	52.3
16	CYP	Cyprus	56	2016	Unicameral	House of Representatives	1.3
17	DNK	Denmark	332	2001–2025	Unicameral	Folketing	6.0
18	EST	Estonia	101	2011, 2015, 2019	Unicameral	Riigikogu	1.4
19	FIN	Finland	121	2003, 2007, 2011	Unicameral	Eduskunta	5.6
20	GBR	UK	832	1997–2017	Lower, Unicameral	House of Commons, National Assembly for Wales, Scottish Parliament	68.6
21	GEO	Georgia	145	2012, 2016	Unicameral	Parliament of Georgia	3.8
22	GGY	Guernsey	39	2016	Unicameral	States	—
23	GRC	Greece	537	2004–2025	Unicameral	Hellenic Parliament, Parliament	10.3
24	GRL	Greenland	38	2014	Unicameral	Inatsisartut	—
25	GTM	Guatemala	152	2012, 2016	Unicameral	Congress	17.9
26	HKG	Hong Kong	55	2012, 2016	Unicameral	Legislative Council	7.5
27	HUN	Hungary	184	2014, 2018	Unicameral	Országgyűlés	9.4
28	IND	India	3332	2014, 2025	Lower, State	Lok Sabha, State Legislature	1441.7
29	IRN	Iran	104	2012, 2016	Unicameral	Majles	89.8
30	ITA	Italy	299	2013, 2018	Upper	Senate	58.6
31	JEY	Jersey	60	2014	Unicameral	States	—
32	KEN	Kenya	253	2013	Lower	National Assembly	56.2
33	KOR	South Korea	251	2012, 2016	Unicameral	National Assembly	51.6
34	LKA	Sri Lanka	221	2015	Unicameral	Parliament	22.1
35	LUX	Luxembourg	60	2013, 2018	Unicameral	Chamber of Deputies	0.7
36	MDA	Moldova	44	2014	Unicameral	Parliament	2.4
37	MKD	Macedonia	101	2014, 2016	Unicameral	Sobranie	1.8
38	MLT	Malta	45	2013	Unicameral	Parliament	0.6
39	NAM	Namibia	69	2015	Lower, Upper	National Assembly, National Council	2.6
40	NGA	Nigeria	200	2015, 2025	Lower, Upper	House of Representatives, Senate	229.2
41	NIC	Nicaragua	84	2012	Unicameral	National Assembly	7.1
42	NLD	Netherlands	144	2012, 2017	Lower	Tweede Kamer	17.9
43	NOR	Norway	174	2025	Unicameral	Storting	5.6
44	NPL	Nepal	264	2014	Unicameral	Constituent Assembly	31.2
45	NZL	New Zealand	117	2008–2017	Unicameral	New Zealand Parliament	5.3
46	PNG	Papua New Guinea	52	2012	Unicameral	National Parliament	10.5
47	PYF	French Polynesia	57	2013	Unicameral	Assembly	0.3
48	ROU	Romania	147	2016	Lower	Chamber of Deputies	18.8
49	RWA	Rwanda	76	2013	Lower	Chamber of Deputies	14.4
50	SGP	Singapore	368	2001–2021	Unicameral	Parliament	6.0
51	SUR	Suriname	57	2015	Unicameral	National Assembly	0.6
52	SVK	Slovakia	164	2006–2016	Unicameral	National Council	5.3
53	SYC	Seychelles	32	2011	Unicameral	National Assembly	0.1
54	TZA	Tanzania	403	2005, 2010, 2015	Unicameral	National Assembly	69.4
55	UGA	Uganda	158	2011	Unicameral	Parliament	49.9
56	URY	Uruguay	119	2015	Lower	Chamber of Deputies	3.4
57	ZAF	South Africa	380	2014	Lower	National Assembly	61.0
58	ZMB	Zambia	66	2011, 2016	Unicameral	National Assembly	21.1
59	ZWE	Zimbabwe	34	2013	Lower	House of Assembly	17.0

Note: One row per country in the analysis sample. Emails counts unique addresses after validation and deduplication. Legislatures with fewer than 30 listed email addresses are excluded. Population is 2024 World Bank total population; it is missing for territories the series does not cover. HIBP indexes only breaches that have become public and been ingested, and we observe at most one address per politician for most of the sample, so every figure is a lower bound.

6,173 people have an observed address and exactly one has two, both of them institutional. The comparison is therefore blocked by address coverage rather than by linkage, and would be identified by collecting one additional address per politician against the Wikidata key. Decade fixed effects are included where legislative start years are available.

2.3 Measuring Exposure to Data Breaches

To evaluate whether politicians’ data has been breached, we check emails against the *Have I Been Pwned* (HIBP) database. Researchers frequently use HIBP to study breaches (see, for, e.g., [Hien et al., 2025](#); [Sood and Cor, 2019](#); [Sood, 2023](#)). For each breach, HIBP reports the type of data exposed, for example, email addresses, hashed passwords, sexual orientation, email messages, etc., and includes metadata such as the year of the breach, its scope (number of records affected), and the industry of the breached online service. HIBP covers breaches dating back to 2007, though publicly available records begin in 2013 ([Table SI1](#) and [Figure SI1](#)).

2.4 Serious Breaches

What makes a breach “serious”? We considered a breached data class *serious* when exposure plausibly causes (i) direct account compromise or recovery abuse across services (a security harm), or (ii) durable, hard-to-reverse risks such as identity theft, financial loss, coercion, discrimination, or persistent tracking (a privacy/pecuniary harm). This aligns with widely used guidance that distinguishes basic identifiers from sensitive PII and “special-category” personal data ([McCallister et al., 2010](#); [European Parliament and Council of European Union, 2016a](#)), recognizes that even apparently low-sensitivity fields can become sensitive when linked (the “mosaic effect”) ([Office of Management and Budget, 2013](#); [McCallister et al., 2010](#)), and treats device/online identifiers as personal data when linkable to a person ([European Parliament and Council of European Union, 2016b](#); [Information Commissioner’s Office, 2024a](#)). In the account-compromise channel, exposure of credentials (includ-

ing recoverable hashed passwords) or recovery factors materially lowers the work factor for *credential stuffing* and related attacks; modern authenticator guidance therefore mandates screening new passwords against known-compromised sets and discourages knowledge-based questions (Grassi et al., 2017; National Institute of Standards and Technology, 2022; OWASP Foundation, 2025; Kanta et al., 2021).

Measurement boundary. *Have I Been Pwned* (HIBP) aggregates public breach corpora and labels each incident with the types of data exposed. However, the public API suppresses some of the most sensitive incidents unless the querier can verify control of the inbox/domain (e.g., certain adult or stigmatizing sites). Consequently, our “serious” counts are *lower bounds* even conditional on observing an address.

Operational categorization. HIBP assigns 146 data classes. We grouped them into three mutually exclusive tiers based on dominant harm mechanism and reversibility:

- **Tier 1 (Profile/Contact).** Names, usernames, email addresses, phone numbers, job titles, IPs, and similar profile attributes. Primary risks are targeting and nuisance; harms are generally indirect and more reversible.
- **Tier 2 (Credential/Account).** Passwords (hashed or clear), password hints, auth tokens, historical passwords, login histories, security Q&A, PINs. Primary risk is cross-site account takeover via credential stuffing or recovery abuse; harms are immediate but usually remediable by rotation/invalidations, with residual risk from unknown reuse (Grassi et al., 2017; OWASP Foundation, 2025; Kanta et al., 2021).
- **Tier 3 (Highly Sensitive).** Financial identifiers (credit cards, CVV, bank accounts), government IDs/passports, biometrics, health/insurance, private messages/email content, browsing histories, photos with sensitive context, SSNs/tax IDs, and similar special-category data. Primary risks are direct fraud, identity takeover, profiling/coercion, and long-term stigma; many elements are non-revocable (IDs/biometrics; leaked com-

munications) and thus create persistent exposure (McCallister et al., 2010; European Parliament and Council of European Union, 2016a; Federal Trade Commission, 2025; National Institute of Standards and Technology, 2020; Information Commissioner’s Office, 2024b).

Table 2. Risk characteristics by tier (mechanism, immediacy, reversibility)

Tier	Dominant mechanism	Attacker cost/time	Immediacy of harm	Reversibility
1: Profile/Contact	Targeting, doxxing, nuisance	Low	Indirect/contingent	High (update/suppress data)
2: Credential/Account	Credential stuffing; recovery abuse; token replay	Low–moderate (automation; offline cracking for weak hashes)	Hours–days (ATO)	Moderate–high (rotate/reset; residual unknown reuse)
3: Highly Sensitive	Payment fraud; identity takeover; profiling/coercion	Low–high (varies by data; fraud tooling exists)	Immediate (financial) and persistent (privacy/identity)	Low (IDs/biometrics/communications hard or impossible to undo)

Implementation and sensitivity analyses. Our primary “serious breach” indicator flagged any incident containing at least one *Tier 2* or *Tier 3* class. We report the resulting rates alongside two pre-specified alternatives—*Serious-Narrow* (Tier 3 only) and *Serious-Password* (password and recovery classes only)—in Table 3; the table reports levels rather than asserting invariance across definitions. The complete primary mapping is listed in Table SI3. This is a formative outcome: the presence of any class defines exposure, and the classes are not interchangeable reflective survey items. We therefore do not use Cronbach’s alpha as a reliability claim; instead, we show how the prevalence changes under substantively distinct class sets.

2.5 Country characteristics

To investigate cross-country variation in exposure, we collect five country-level predictors. To measure a country’s level of digitization, we collect the UN e-Government Development

Table 3. Sensitivity of serious-breach prevalence to outcome definition

Outcome definition	Official rate (N)	Commercial rate (N)	Official, role excluded	Commercial, role excluded
Primary (Tier 2+3)	17.2			
Narrow (Tier 3 only)	1.8			
Password-related	16.6			

Note: Rates are proportions of email addresses. The primary definition includes Tier 2 and Tier 3 classes; the narrow definition includes Tier 3 only; the password definition includes credential and recovery classes. Role-account columns exclude local-part prefixes commonly used for shared institutional mailboxes.

Index (EGDI), a composite of government online services, telecommunication infrastructure, and human capital (United Nations Department of Economic and Social Affairs, 2020; Teorell et al., 2026). Regime type comes from the V-Dem electoral democracy index over the same window (Coppedge et al., 2026). For national cybersecurity effort, we use the 2020 edition of the Global Cybersecurity Index (GCI), a 0–100 score of national cybersecurity commitments (International Telecommunication Union, 2021). We code whether English is an official language, taken from the CEPII GeoDist database (Mayer and Zignago, 2011). Finally, we collect countries’ GDP per capita (World Bank, 2025). Where multi-year data exist, we take the average over 2010 to 2020—for digitization this covers the five biennial editions of the survey published between 2012 and 2020. The UN indices cover member states only, which excludes several small territories (Bermuda, French Polynesia, Greenland, Guernsey, Hong Kong, and Jersey), and Andorra lacks a V-Dem score, giving 52 countries in the final analytical sample.

3 Results

3.1 Prevalence of Politician Email Breaches

Nearly one in three observed email addresses is linked to at least one public breach (Panel A, Table 4). This pools incidents that differ in what they imply about the account holder, and Table SI6 decomposes it: 23.5% of addresses appear only because a data broker or scraper held them, while 15.9% appear in a breach of a service the politician plausibly held an account with. Rates below should be read as exposure of the address, not as compromise

of the politician’s accounts. However, there is a sharp skew in the distribution. While the mean number of breaches per account is 1 ($\hat{\sigma} = 3.2$), the median is 0, with 19.5% of accounts breached multiple times, some as many as hundreds of times.

Serious data breaches were associated with 21.6% of observed email addresses (Panel B, Table 4). Of the breached addresses, the table reports the corresponding serious-breach share. Again, we see a substantial skew. Most addresses are never seriously breached, but a smaller proportion are breached multiple times.

Table 4. Number of breaches per politician email address

	N	Mean	SD	Min	P25	Median	P75	Max	≥ 1	≥ 2
<i>Panel A: All breaches</i>										
All emails	12,385	1.0	3.2	0	0	0	1	256	33.0%	19.5%
Government emails	9,371	0.8	1.7	0	0	0	1	18	28.4%	16.1%
Personal emails	3,014	1.8	5.6	0	0	0	2	256	47.3%	30.3%
<i>Panel B: Serious breaches</i>										
All emails	12,385	0.5	2.1	0	0	0	0	189	21.6%	10.6%
Government emails	9,371	0.3	0.9	0	0	0	0	9	17.2%	7.6%
Personal emails	3,014	1.0	4.0	0	0	0	1	189	35.2%	20.0%

Note: The unit is an email address, not a politician; politicians with more than one listed address contribute more than one row. The final two columns give the share of addresses appearing in at least one and at least two distinct breaches. A breach is coded serious if it exposed at least one of the 38 data classes listed in Table S13. HIBP indexes only breaches that have become public and been ingested, and we observe at most one address per politician for most of the sample, so every figure is a lower bound.

Table 5. Data classes exposed in breaches involving politicians

Data class	N	%	Data class	N	%	Data class	N	%
1 Email addresses	4092	100.0%	21 Marital statuses	99	2.4%	41 Payment histories	38	0.9%
2 Names	3481	85.0%	22 Religions	99	2.4%	42 Survey results	34	0.8%
3 Phone numbers	3202	78.2%	23 Email messages	96	2.3%	43 User website URLs	31	0.8%
4 Job titles	2842	69.4%	24 Password hints	81	2.0%	44 Telecommunications carrier	30	0.7%
5 Physical addresses	2703	66.0%	25 Ethnicities	77	1.9%	45 Nationalities	27	0.7%
6 Social media profiles	2645	64.6%	26 Home ownership statuses	77	1.9%	46 Private messages	27	0.7%
7 Passwords	2598	63.5%	27 Auth tokens	75	1.8%	47 Relationship statuses	21	0.5%
8 Geographic locations	2090	51.1%	28 Occupations	75	1.8%	48 Company names	20	0.5%
9 Employers	1868	45.6%	29 PINs	69	1.7%	49 Deceased statuses	18	0.4%
10 Genders	1714	41.9%	30 Partial credit card data	67	1.6%	50 Website activity	16	0.4%
11 Dates of birth	1594	38.9%	31 Credit status information	64	1.6%	51 Professional skills	15	0.4%
12 IP addresses	1574	38.5%	32 Family structure	61	1.5%	52 Credit cards	14	0.3%
13 Usernames	1004	24.5%	33 Financial investments	61	1.5%	53 Passport numbers	9	0.2%
14 Salutations	419	10.2%	34 Net worths	61	1.5%	54 Profile photos	8	0.2%
15 Education levels	414	10.1%	35 Personal interests	61	1.5%	55 Support tickets	8	0.2%
16 Purchases	246	6.0%	36 Government issued IDs	55	1.3%	56 Cryptocurrency wallet addresses	5	0.1%
17 Spoken languages	220	5.4%	37 Device information	50	1.2%	57 Social security numbers	5	0.1%
18 Political donations	154	3.8%	38 Browser user agent details	43	1.1%	58 Account balances	4	0.1%
19 Income levels	134	3.3%	39 Bios	41	1.0%	59 Health insurance information	4	0.1%
20 Job applications	113	2.8%	40 Bank account numbers	40	1.0%	60 Loyalty program details	4	0.1%

Note: Percentages are of breached addresses, not of all addresses, so they do not sum to 100. Read in three column blocks, ranked by frequency. HIBP indexes only breaches that have become public and been ingested, and we observe at most one address per politician for most of the sample, so every figure is a lower bound.

The most common data exposed in breaches involving politicians are email addresses, names, phone numbers, job titles, and physical addresses (Table 5). While these are mostly

innocuous data, passwords are the most common serious data class exposed. More than 63% of breached addresses had a password-related class exposed; this denominator is breached addresses, not all observed addresses. In nearly 1.6% of the account breaches, credit card information was revealed. Bank account numbers were compromised in 1.0% of the breaches, and government-issued IDs in 1.3%. Some breaches also exposed private communications, such as emails (2.3%) and other private messages (0.7%).

3.2 Official Vs. Personal Emails

[Table 4](#)’s numbers paint too rosy a picture. As we note above, for many politicians, we only have their official government email addresses. To shed light on the issue, we analyze how the number of breaches varies by official vs. personal emails.

[Table 4](#) tabulates the number of breaches by personal and official email accounts. Of the 12,385 observed addresses, 3,014 are personal. A personal address is almost twice as likely as an official one to appear in at least one breach ([Table 4](#)). The breach rate for official emails is 28.4% vs. 47.3% for personal emails (Panel A, [Table 4](#)). The relative difference for serious data breaches is more pronounced, with the respective numbers being 17.2% and 35.2% (Panel B, [Table 4](#)). It is tempting to read the personal-address rate as the truer measure of a politician’s exposure, but two considerations argue against doing so directly. The next subsection shows the difference may partly reflect selection into publishing a personal address, and the provenance decomposition in [Table SI6](#) shows that much of the official-address exposure is broker aggregation of addresses published by design rather than compromise of accounts.

3.3 Who Gets Breached?

The personal vs. official email results may be driven by selection bias. Politicians who use personal email providers may be more lax about their security or anticipate shorter political stints. To better address the confounds and to study the variation in exposure more generally,

we model breaches as a function of politician characteristics. Given the skew in the data (Table 4), we model the likelihood of an email breach rather than the number of breaches. In all, we estimate the following equation:

$$\begin{aligned} \text{Breach}_{i,t} = & \beta_1 \text{PersonalEmail}_i + \beta_2 \text{Female}_i + \beta_3 \text{SocialMedia}_i \\ & + \gamma_{c(i)} + \tau_t + \theta_i + \lambda_{c(i),i} + \varepsilon_{i,t}, \quad (1) \end{aligned}$$

where i indexes addresses, c indexes country, and t indexes the decade of the legislative start year. The data are cross-sectional, with one observation per address; t enters only through decade fixed effects. Our base specification models the risk of a breach (or serious breach) as a function of email type (personal vs. official), country, and decade-fixed effects (based on the legislative start year). For the EveryPolitician data, we have more variables available. We take advantage of the richer data by adding controls for politician gender (β_2), whether or not the politician has a known Twitter or Facebook account (β_3), legislature type (lower house, upper house, unicameral legislature, $\lambda_{c(i),i}$), and political party (θ_i). Since we expect observations within the same country to share unobserved characteristics that may induce correlation in the error terms, we cluster the standard errors at the country level. Because the design has only 59 country clusters and one country supplies 26.9% of addresses, we also report null-imposed WCR11 wild-cluster p-values with 9,999 Rademacher draws as a small-cluster diagnostic. We estimate the models using *fixest* in R (Bergé, 2018).

Before turning to the results, a necessary caveat: because of potential confounds, cross-group differences cannot be interpreted as stemming from some essential aspect of the group. For instance, one of the many potential confounds that could vitiate comparisons is that the risk of exposure is correlated with political tenure. If, for instance, male legislators may have longer tenures than female legislators, the coefficients on gender will partly reflect differences in tenure (which is not modeled).

Across all specifications, personal emails are strongly linked to a higher probability

Table 6. Linear probability models of breach exposure

	dbreach		dbreach_serious	
	(1)	(2)	(3)	(4)
Personal email	0.286*** (0.016)	0.240*** (0.030)	0.264*** (0.012)	0.295*** (0.041)
Female		0.009 (0.012)		0.023 ⁺ (0.012)
Social media		0.087*** (0.021)		0.079*** (0.015)
# Country	59	53	59	53
# Decade	4	3	4	3
# Legislature type	—	3	—	3
# Political party	—	359	—	359
Dependent variable mean	0.330	0.387	0.216	0.245
Observations	12,385	7,082	12,385	7,082
R ²	0.303	0.511	0.276	0.459
Country fixed effects	✓	✓	✓	✓
Decade fixed effects	✓	✓	✓	✓
Legislature type fixed effects		✓		✓
Political party fixed effects		✓		✓

Note: Linear probability models estimated with `fixest::feols`. The unit is an email address. Standard errors, in parentheses, are clustered on country; note that 59 country clusters are dominated by India, which supplies 26.9% of the sample, so the effective number of clusters is far smaller than the nominal one. Columns 2 and 4 are estimated on the EveryPolitician subsample only, since gender, party and social-media fields exist only there; they are a different sample, not a robustness check on columns 1 and 3. Social media indicates that EveryPolitician recorded a Twitter or Facebook handle, which is partly a measure of how completely the record is populated. A breach is coded serious if it exposed at least one of the 38 data classes listed in [Table SI3](#).

⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

of breach (see [Table 6](#)). Personal emails have a 24–30 percentage points greater chance of having a breach (serious breach) associated with them ($p < .001$, [Table 6](#) Columns (1)–(4)). In all, given that personal emails better reflect online behavior ([Section 2](#)), we believe a more realistic estimate of the percentage of politicians with a serious breach is over 50%.

Next, we describe the variation in the likelihood of being involved in a breach by other characteristics of the politician. Adjusting for the other covariates, female politicians are more likely to have a serious breach than male politicians, with a 2.3 percentage point higher probability ($p < .1$, [Table 6](#) Column (4)). The kinds of politicians for whom we have public social media profiles are likelier to have had their data breached. There is a 7.8 percentage point increase in the likelihood of a serious breach ($p < .001$, [Table 6](#) Column (4)).

3.4 Country-Level Predictors

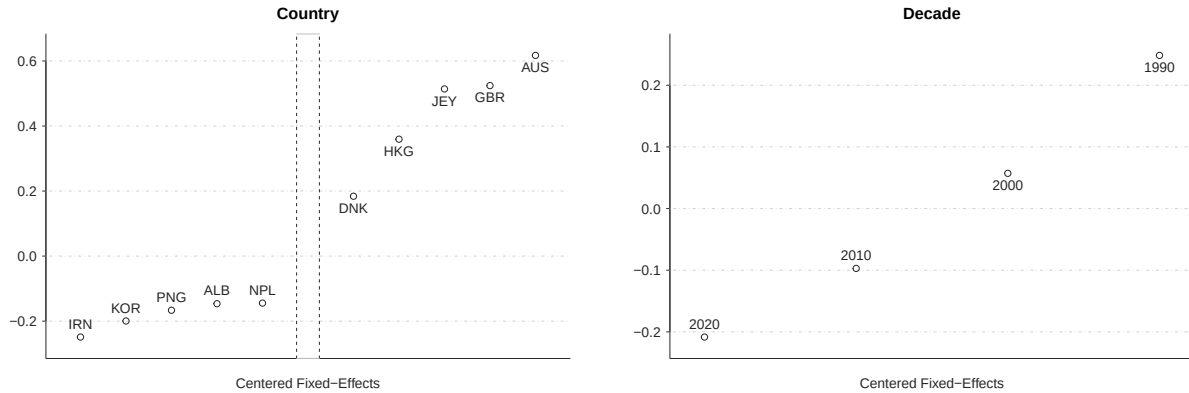


Figure 1. Country and decade mean-centered fixed effects of the probability of a serious breach. This figure reports the top five and bottom five estimated fixed effects for the country (left panel) and the fixed effects for a decade (right panel) from Model (3) in [Table 6](#). All plotted effects are centered such that the mean is zero. See [Table SI5](#) for all countries’ estimated effects.

[Figure 1](#) reports the mean-centered fixed effects for countries with the highest and lowest breach rates conditional on the email type and time fixed effects. The top five countries with the highest probability of having accounts breached are Australia, the UK, Jersey, Hong Kong, and Denmark. The bottom five are Iran, Korea, Papua New Guinea, Albania, and

Nepal. [Figure 1](#) (right panel) also confirms that older accounts have higher probabilities of breach.

One misreading of the above would be that countries at the top of the rankings are those with lax security infrastructures, while those at the bottom are better defended. Here we test an alternative explanation, that exposure tracks integration into the global digital-services economy more than it tracks regime quality or security investment. We regress the estimated country fixed effects on the five covariates. Consistent with the top-ranked countries ([Table SI5](#)), income alone is a statistically significant predictor and explains a substantial share of the variation ($R^2 = 0.36$ for any breach, 0.24 for serious breaches). However, its coefficient becomes zero once the other covariates are included in the model ([Table SI4](#)). Instead, having English as an official language predicts 21 (17) percentage points higher exposure to any (serious) breach ($p < .001$). Electoral democracy predicts higher exposure to any breach ($p < .001$), and to serious breaches ($p = .007$). Digitization is positive but does not reach the 5% level for serious breaches with HC3 inference ($p = .079$). The GCI coefficient is close to zero for both outcomes, so we find no evidence that its national cybersecurity-commitment score predicts the exposure of political elites. [Figure 2](#) splits the explained variance into each covariate’s Shapley share, reaching similar conclusions. For serious breaches, having English as the official language (0.17 of the total 0.50) and digitization (0.11) contribute the largest shares, and the GCI the least.

3.5 Which Breaches Are Most Responsible?

[Table 7](#) reports the most common data breaches involving politicians. The two most common breaches involving politicians are the *db8151dd* (2020) and the *OnlinerSpambot* (2017) breaches. The *db8151dd* breach, originating from Covve (a Customer Relationship Management app), leaked fairly benign information, such as email addresses, job titles, names, phone numbers, physical addresses, and social media profiles. (The same six data types are also the six most common kinds of data that are compromised in breaches more generally

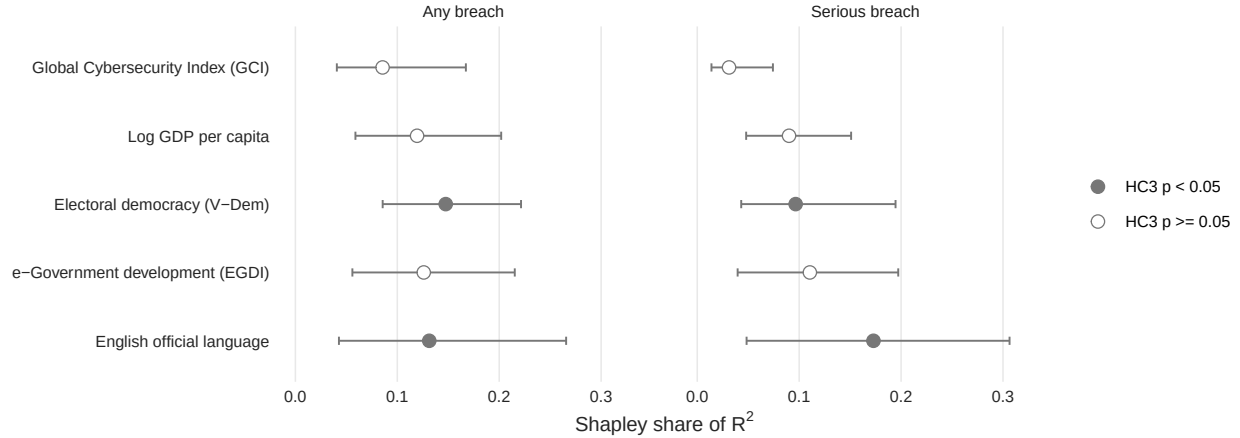


Figure 2. Decomposition of the cross-country variation in breach exposure. Each marker is a covariate’s (Shapley) share of the explained variance in the estimated country-level aggregates. Capped horizontal lines are 95% intervals from 10000 bootstrap resamples of countries. Covariates whose coefficient in the full regression (Table SI4) reaches significance at the 5% level are shown with darker markers.

Table 7. Breach incidents most often involving politician email addresses

Breach	N	%	Domain	Breach date	Added date	Lag	Accounts	Data classes	Sensitive
1 db8151dd	1441	35.2%	covve.com	2020-02-20	2020-05-15	0.2 years	22.8M	6	
2 OnlinerSpambot	1231	30.1%	—	2017-08-28	2017-08-29	0.0 years	711.5M	2	✓
3 PDL	1128	27.6%	—	2019-10-16	2019-11-22	0.1 years	622.2M	7	
4 VerificationsIO	1037	25.3%	verifications.io	2019-02-25	2019-03-09	0.0 years	763.1M	10	
5 LinkedIn	489	11.9%	linkedin.com	2012-05-05	2016-05-21	4.0 years	164.6M	2	✓
6 LinkedInScrape	385	9.4%	linkedin.com	2021-04-08	2021-10-02	0.5 years	125.7M	7	
7 Apollo	340	8.3%	apollo.io	2018-07-23	2018-10-05	0.2 years	125.9M	8	
8 Intelimost	294	7.2%	intelimost.com	2019-03-10	2019-04-02	0.1 years	3.1M	2	✓
9 Cit0day	271	6.6%	cit0day.in	2020-11-04	2020-11-19	0.0 years	226.9M	2	✓
10 Twitter200M	269	6.6%	twitter.com	2021-01-01	2023-01-05	2.0 years	211.5M	4	
11 Collection1	256	6.3%	—	2019-01-07	2019-01-16	0.0 years	772.9M	2	✓
12 ExploitIn	233	5.7%	—	2016-10-13	2017-05-06	0.6 years	593.4M	2	✓
13 AntiPublic	226	5.5%	—	2016-12-16	2017-05-04	0.4 years	458.0M	2	✓
14 DemandScience	205	5.0%	demandscience.com	2024-02-28	2024-11-13	0.7 years	121.8M	7	
15 Gravatar	196	4.8%	gravatar.com	2020-10-03	2021-12-05	1.2 years	114.0M	3	
16 TelegramCombolist	164	4.0%	—	2024-05-28	2024-06-03	0.0 years	361.5M	3	✓
17 Nitro	160	3.9%	gonitro.com	2020-09-28	2021-01-19	0.3 years	77.2M	3	✓
18 FairVoteCanada	154	3.8%	fairvote.ca	2024-03-02	2024-10-21	0.6 years	0.1M	5	
19 YouveBeenScraped	147	3.6%	—	2018-10-05	2018-12-06	0.2 years	66.1M	6	
20 NotSOCRadar	129	3.2%	—	2024-08-03	2024-08-09	0.0 years	282.5M	1	
21 Dropbox	128	3.1%	dropbox.com	2012-07-01	2016-08-31	4.2 years	68.6M	2	✓
22 Canva	127	3.1%	canva.com	2019-05-24	2019-08-09	0.2 years	137.3M	5	✓
23 IIMJobs	113	2.8%	iimjobs.com	2018-12-31	2021-05-21	2.4 years	4.2M	9	✓
24 RiverCityMedia	105	2.6%	rivercitymediaonline.com	2017-01-01	2017-03-08	0.2 years	393.4M	4	
25 ShareThis	100	2.4%	sharethis.com	2018-07-09	2019-03-03	0.6 years	41.0M	4	✓

Note: Ranked by the number of politician addresses in the breach. Percentages are of breached addresses. Lag is the interval between the breach date HIBP records and the date HIBP added it; because the breach date is often HIBP’s approximation and the added date reflects ingestion rather than disclosure, the lag is an ingestion latency, not a disclosure latency. HIBP indexes only breaches that have become public and been ingested, and we observe at most one address per politician for most of the sample, so every figure is a lower bound.

Table 5.)

OnlinerSpambot was a more serious breach. The *OnlinerSpambot* incident was not just another breach but a live cybercrime operation, with the stolen email access weaponized to spread malware that steals bank account credentials and credit card details (Hayashi, 2017;

(Kelion, 2017; Trend Micro, 2017; Have I Been Pwned, 2018). Both email addresses and passwords were also released.

The *OnlinerSpambot* breach was detected within a day. Other serious breaches like the *LinkedIn* breach, which is the fifth-most common and leaked email addresses and passwords, went undetected for four full years (Column (7), Table 7) until the leaked data went on sale on a dark market site (Have I Been Pwned, 2018). Some breaches in HIBP took as long as 12 years before the public found out (Table SI1). Notable, many breaches such as *OnlinerSpambot*, unlike *db8151dd* and *LinkedIn*, are not tied to any single online service or domain, and it is still unknown where bad actors sourced the credentials from (Kelion, 2017; Trend Micro, 2017; Have I Been Pwned, 2018).

3.6 Leaks to Floods? Breaches Over Time

Finally, we investigate how the exposure from data breaches grows over time. We start with the 9,200 email accounts where legislative start dates are available. Figure 3 plots the percentage of emails with a breach associated with them over time. The middle and bottom panel of Figure 3 plots the total number of politician emails and breach incidents, respectively.

The breach rate starts at zero, but by January 2025, it reaches 37.1%. This is higher than the 33.0% reported in Table 4 (Panel A) because the two are computed on different samples: the series here is restricted to the 9,200 accounts with a datable legislative start, whereas Panel A uses all 12,385. The excluded accounts are almost entirely the 2025 cohort, which has not yet entered the risk set and is largely unbreached. While the bottom panel of Figure 3 shows a gradual increase in publicly-known breaches over time, breach rates do not follow that pattern. Instead, increases in exposure are concentrated around a few incidents, as indicated by many sharp stepwise increases in breach rate among extant politician emails. Five incidents with an outsized impact on politician breaches are the *LinkedIn* (2012), *Dropbox* (2012), *OnlinerSpambot* (2017), *VerificationsIO* (2019), and *db8151dd* (2020) breaches

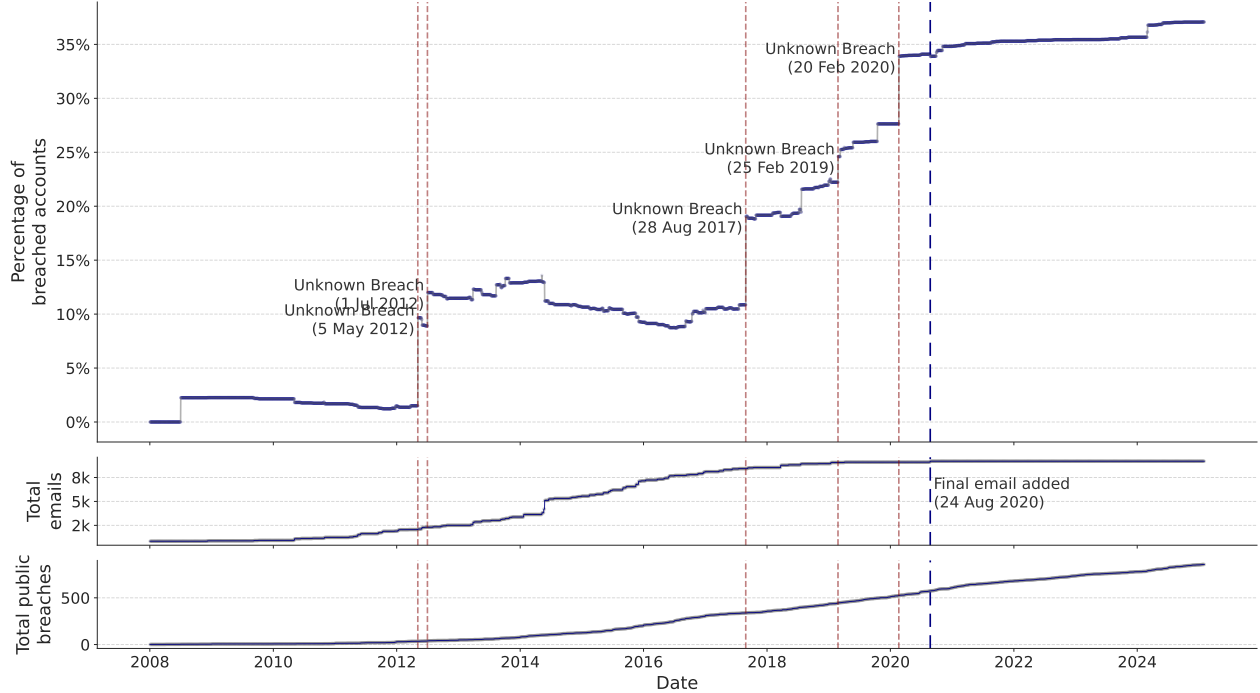


Figure 3. Breach rate of politician emails over time. This figure plots the percentage of breached politician email accounts ($n = 9,200$) based on the cumulative number of extant email accounts (based on the observed earliest legislative start date) and publicly-known data breaches. Each point represents the percentage of total extant emails found in breaches as of that date. The top panel annotates the five largest percentage jumps and the corresponding breach incidents (dashed vertical red lines). *Disqus* and *Dropbox* happened the same day. The middle panel tracks the cumulative number of emails. The bottom panel tracks the cumulative number of data breaches.

(see also Table 7). Overall, Figure 3 suggests that a small number of large-scale breaches drive exposure.²

4 Discussion

Data breaches expose people to theft, extortion, impersonation, and blackmail. When the target is a politician, the social cost of a breach is likely considerably higher. Politicians handle sensitive government information, influence policy, and engage with influential fig-

²The small dips in the percentage of accounts with a breach reflect those cases where a new cohort of emails joins the pool. Figure SI3 follows a small and fixed cohort of emails that existed before 2007 and shows a monotonic pattern that is similar to the pattern we see in Figure 3, with a few incidents driving most of the breaches, and with breach rate reaching close to 70% by Jan 2025. This aligns with Figure 1 (and Figure SI2), confirming that earlier email accounts have a higher breach probability.

ures. Their public statements shape discourse. The corresponding risks from impersonation, blackmail, etc., are hence greater (for example, [Harding, 2016](#); [Almasy, 2017](#); [BBC News, 2019, 2020](#); [Witman and Mackelprang, 2022](#)), and compound the substantial cybersecurity gaps documented in government infrastructure worldwide ([Ribeiro et al., 2025](#)). The threat is amplified when we account for systematic efforts to exploit information or election interference, misinformation campaigns, and undermining of democracy ([Harding, 2016](#); [CrowdStrike, 2016](#); [Almasy, 2017](#); [BBC News, 2017, 2019](#); [Shires, 2019](#); [Bizga, 2020](#); [Stahie, 2020](#); [BBC News, 2020](#); [Tomz and Weeks, 2020](#)).

In this study, we assemble a large database of politician email addresses to assess exposure in public breach corpora. Conservatively, more than one in five observed addresses have had sensitive data breached at least once. Because addresses are an incomplete proxy for people and HIBP is incomplete, this should not be read as a population-level rate for politicians.

The composition of that exposure matters as much as its level. Most of it arises because a third party held the address rather than because an account was compromised, and only 2.32% of addresses appear in corpora derived from infostealer malware, the one channel that indicates a compromised device. Benchmarked against a general-population sample on a shared catalogue and window, politician addresses appear in breaches several times less often, not more. We do not read that as evidence that politicians are safer: their published addresses are institutional and comparatively recent, and the samples differ in national coverage of breach corpora. We do read it as evidence that a bare prevalence figure, uncomparing and undecomposed, cannot support claims about how exposed political elites are relative to anyone else.

There is a sharp skew in exposure, with a small proportion of addresses repeatedly compromised. This pattern could arise from several reasons, including credential reuse, weak passwords, and attacks targeting certain politicians ([Harding, 2016](#); [Almasy, 2017](#); [BBC News, 2019](#)).

Correspondingly, there is also a sharp skew in the number of politicians entangled in each breach. A few large breaches account for most of the exposure. This pattern mirrors that of the general population (Sood and Cor, 2019). Part of the pattern is founded in leaks from a few widely used online services, such as LinkedIn and Dropbox.

Prior studies of data breaches have focused on the general population. Using the same HIBP database, a study of a representative sample of 5000 American adults found breach rates of 80% (Sood and Cor, 2019), a smaller online survey of 413 participants found a breach rate of 73% (Mayer et al., 2021), and one using over a million Florida registered voters found 84% (Sood, 2023). In comparison, our estimated prevalence of serious breaches of politicians exceeds 50%. These numbers suggest that politicians are not much better protected than the public despite the higher stakes of compromise. The harm does not stop with the individual politician either. Breached data feeds hack-and-leak operations (Shires, 2019). And in survey experiments, people who learn that an election was interfered with come away trusting democracy less (Tomz and Weeks, 2020). The breached politician data we document here plausibly lowers the cost of mounting such compromising operations, at least by reducing the effort needed to obtain compromising material in the first place.

4.1 Real-world breaches

Political data breaches, though rarely documented systematically (Dunn Cavelty and Wenger, 2020), as in this study, are far from hypothetical. Recent years have seen many high-profile incidents of breaches that vividly illustrate the potential fallout when elected officials are compromised (Harding, 2016; CrowdStrike, 2016; Almasy, 2017; BBC News, 2017, 2019; Bizga, 2020; Stahie, 2020; BBC News, 2020; Witman and Mackelprang, 2022).

For instance, in the lead-up to the 2016 US presidential election, Russian-affiliated actors compromised the email systems of the Democratic National Committee and senior Democrats. The leaked emails were widely disseminated online and fueled disinformation campaigns and political attacks, undermining public trust in a deeply polarized election

([Harding, 2016](#); [CrowdStrike, 2016](#)). A year later, in a parallel case, Emmanuel Macron’s presidential campaign was hacked just days before the French runoff vote. Thousands of authentic internal emails and documents, along with fabricated materials, were leaked in an apparent attempt to compromise his candidacy ([Almasy, 2017](#); [Shires, 2019](#)). In 2019, personal data belonging to hundreds of German politicians, including Chancellor Angela Merkel and others from major political parties, was leaked, exposing contact details, private chats, ID documents, and credit card details ([BBC News, 2019](#)). In 2020, attackers successfully accessed Twitter’s internal tools through social engineering, hijacking the verified accounts of prominent figures such as Barack Obama, Joe Biden, and Michael Bloomberg, among others. The breach was used to promote a Bitcoin scam that netted over US\$100,000 in cryptocurrency before getting shut down ([BBC News, 2020](#); [Witman and Mackelprang, 2022](#)). More recently, in 2025, it was revealed that the US Director of National Intelligence, Tulsi Gabbard, had repeatedly reused the same weak password across multiple personal accounts, raising concerns about the cybersecurity practices of top intelligence officials ([Greenberg, 2025](#)).

4.2 Implications for practice

These incidents, along with the broader patterns we document, highlight the need for stronger safeguards surrounding public officials and political digital infrastructure ([Ribeiro et al., 2025](#)). While some breaches involved sophisticated tactics in attacking third-party services ([BBC News, 2020](#); [Witman and Mackelprang, 2022](#)), many arguably stem from preventable vulnerabilities, such as weak passwords, credential reuse, and the absence of two-factor authentication ([Consolvo et al., 2021](#); [BBC News, 2019](#); [Connolly, 2019](#); [BBC News, 2020](#); [Greenberg, 2025](#); [Cooper, 2025](#); [Ribeiro et al., 2025](#)). Interviews with US campaign staff confirm that political campaigns routinely lack dedicated IT support, rely on personal devices and accounts for sensitive work, and adopt security measures on an ad-hoc basis ([Consolvo et al., 2021](#)). At a minimum, governments and political organizations should institutionalize

baseline levels of cybersecurity hygiene.

Before listing those levers, we note what they cannot reach. The provenance decomposition in [Table SI6](#) shows that the single largest channel of exposure is commercial aggregation: addresses harvested into broker, scrape and enrichment databases that the politician never transacted with. Password managers and multi-factor authentication do nothing about this. Reducing it requires limiting the bulk publication and resale of official contact details, which is a procurement and data-protection question rather than a security-hygiene one. The levers below therefore target the service-compromise and malware channels, which are smaller but are where individual and institutional practice binds.

Given our measurement scope, the most effective levers align with the Tier 2/3 risk channels (credential compromise and exposure of highly sensitive attributes): (i) *Eliminate credential reuse* via mandatory password managers and screening new passwords against known-compromised sets; pair breach notifications with just-in-time prompts, which have been shown to increase password changes ([Albayram and Walker, 2024](#)), and use password meters as a visual aid ([Ur et al., 2012](#)); (ii) *Harden authentication* by enforcing phishing-resistant multi-factor authentication on official domains (e.g., security keys/platform authenticators); as precedent, the UK Parliament tightened authentication controls after the 2017 incident ([BBC News, 2017](#); [UK Parliament, 2017](#)); (iii) *Monitor and respond at the domain boundary*: centrally track verified government domains against breach corpora, and when an official address appears in a newly added incident, notify the holder, rotate credentials, and revoke third-party tokens relying on that address; (iv) *Make the secure action the default*: deliver short, role-specific training embedded in these workflows rather than standalone awareness courses. These steps target the concrete downstream risks we observe (credential-stuffing, high-fidelity targeting) without over-inferring national-security impacts we do not measure.

4.3 Limitations

First, addresses are a lower-bound instrument: we miss cases where politicians use unobserved personal addresses, and HIBP suppresses certain sensitive incidents via its public API. Second, some official boxes are role accounts; some personal/campaign domains have professional IT support—classification errors would attenuate differences. Third, not all breaches are public; lags between breach and public availability (median ≈ 0.5 years) bias us toward undercounting recent exposure. Fourth, we do not measure compromise of classified networks; our findings concern public breach corpora tied to the address. Fifth, cross-sectional comparisons (e.g., gender or country FE) may reflect correlated behaviors such as tenure, media visibility, or differential address publication rather than inherent susceptibility. Finally, the provenance classification is our own and rests on breach descriptions rather than on forensic evidence of how each corpus was assembled; several large corpora explicitly mix stealer output with recycled credential lists, so the boundary between the combolist and service classes is not sharp. The population benchmark is subject to a stronger caveat still: it compares samples drawn from different countries, and because national representation in breach corpora is uneven, the level difference cannot be attributed to differences in behavior or security practice.

4.4 Conclusion

In summary, politicians’ contact data circulates widely in public breach corpora, and the sensitive-data share is large enough to warrant stronger security practices for public officials. But the decomposition above should redirect where that effort goes. The largest single channel is not weak passwords; it is commercial aggregation of official contact details, which no amount of individual hygiene addresses and which is a matter for procurement and data-protection enforcement. The credential-hygiene measures we recommend remain justified, and they target the smaller service-compromise and malware channels where they can actually bind.

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Supplementary Information

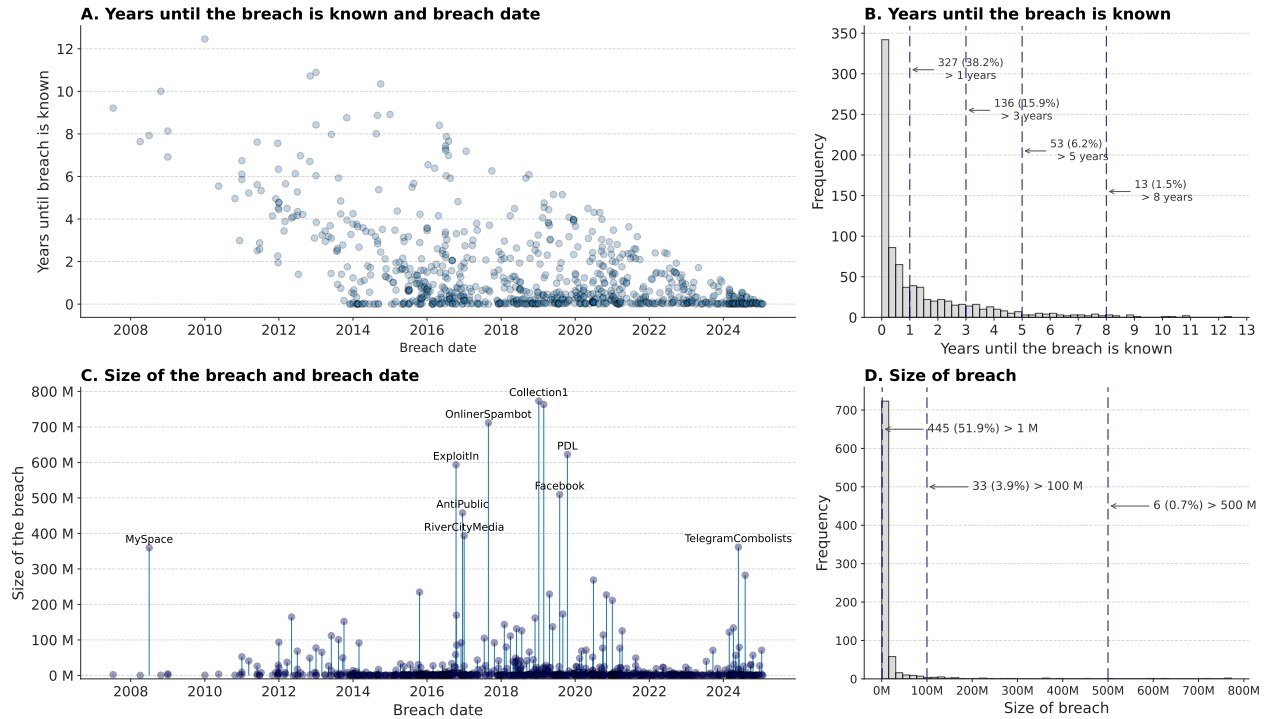


Figure SI1. Summary of data breaches recorded in HIBP. The subfigures summarize the time lag between the breach occurrence and its public disclosure on HIBP (*Years until the breach is known*) and the number of accounts compromised (*Size of breach*).

Table SI1. Summary of HIBP breaches ($n = 857$)

	Mean (1)	Std dev (2)	Min (3)	25p (4)	Median (5)	75p (6)	Max (7)
Breached accounts (in 100,000s)	169.59	677.67	0.01	2.7	11.29	58.15	7729.05
Number of compromised data types	5.29	2.76	1.0	4.0	5.0	7.0	25.0
Breach date (yyyy-mm-dd)	2018-10-11	—	2007-07-12	2016-02-19	2018-12-26	2021-04-23	2025-01-24
Date added to HIBP (yyyy-mm-dd)	2020-02-21	—	2013-11-30	2017-05-16	2020-02-20	2023-01-02	2025-02-02
Years till breach is public on HIBP	1.36	1.95	0.0	0.06	0.51	1.94	12.46

Note: Years till the breach is public on HIBP is the lapse between the date of the breach and the date added to the HIBP repository.

Table SI2. Types of data exposed in breaches ($n = 857$)

1–50			51–100			101–146		
Data type	#	(%)	Data type	#	(%)	Data type	#	(%)
1 Email addresses	851	99.3%	51 Vehicle details	6	0.7%	101 Cellular network names	1	0.1%
2 Passwords	618	72.1%	52 Credit status information	5	0.6%	102 Charitable donations	1	0.1%
3 Usernames	431	50.3%	53 Family members' names	5	0.6%	103 Citizenship statuses	1	0.1%
4 Names	428	49.9%	54 Historical passwords	5	0.6%	104 Clothing sizes	1	0.1%
5 IP addresses	353	41.2%	55 Home ownership statuses	5	0.6%	105 Comments	1	0.1%
6 Phone numbers	276	32.2%	56 Partial dates of birth	5	0.6%	106 Company names	1	0.1%
7 Dates of birth	234	27.3%	57 Personal health data	5	0.6%	107 Customer feedback	1	0.1%
8 Physical addresses	218	25.4%	58 Social connections	5	0.6%	108 Customer interactions	1	0.1%
9 Genders	164	19.1%	59 User website URLs	5	0.6%	109 Deceased date	1	0.1%
10 Geographic locations	121	14.1%	60 Age groups	4	0.5%	110 Deceased statuses	1	0.1%
11 Website activity	77	9.0%	61 Chat logs	4	0.5%	111 Delivery instructions	1	0.1%
12 Purchases	66	7.7%	62 Ages	3	0.4%	112 Device serial numbers	1	0.1%
13 Social media profiles	49	5.7%	63 Nicknames	3	0.4%	113 Device usage tracking data	1	0.1%
14 Private messages	37	4.3%	64 Personal descriptions	3	0.4%	114 Eating habits	1	0.1%
15 Job titles	30	3.5%	65 Photos	3	0.4%	115 Employment statuses	1	0.1%
16 Partial credit card data	29	3.4%	66 SMS messages	3	0.4%	116 Encrypted keys	1	0.1%
17 Employers	27	3.2%	67 Sexual fetishes	3	0.4%	117 Fitness levels	1	0.1%
18 Browser user agent details	22	2.6%	68 Browsing histories	2	0.2%	118 Flights taken	1	0.1%
19 Device information	22	2.6%	69 Credit card CVV	2	0.2%	119 HIV statuses	1	0.1%
20 Salutations	20	2.3%	70 Cryptocurrency wallet addresses	2	0.2%	120 Licence plates	1	0.1%
21 Government issued IDs	18	2.1%	71 Driver's licenses	2	0.2%	121 Living costs	1	0.1%
22 Spoken languages	18	2.1%	72 Drug habits	2	0.2%	122 Login histories	1	0.1%
23 Marital statuses	15	1.8%	73 Financial investments	2	0.2%	123 Loyalty program details	1	0.1%
24 Security questions and answers	13	1.5%	74 Financial transactions	2	0.2%	124 MAC addresses	1	0.1%
25 Bios	12	1.4%	75 Health insurance information	2	0.2%	125 Mnemonic phrases	1	0.1%
26 Education levels	12	1.4%	76 Homepage URLs	2	0.2%	126 Mothers maiden names	1	0.1%
27 Income levels	12	1.4%	77 IMEI numbers	2	0.2%	127 Parenting plans	1	0.1%
28 Profile photos	12	1.4%	78 IMSI numbers	2	0.2%	128 Password strengths	1	0.1%
29 Physical attributes	10	1.2%	79 Loan information	2	0.2%	129 Payment methods	1	0.1%
30 Account balances	9	1.1%	80 Net worths	2	0.2%	130 Places of birth	1	0.1%
31 Auth tokens	9	1.1%	81 PINs	2	0.2%	131 Purchasing habits	1	0.1%
32 Ethnicities	9	1.1%	82 Partial phone numbers	2	0.2%	132 Races	1	0.1%
33 Nationalities	9	1.1%	83 Password hints	2	0.2%	133 Recovery email addresses	1	0.1%
34 Payment histories	9	1.1%	84 Personal interests	2	0.2%	134 Reward program balances	1	0.1%
35 Religions	9	1.1%	85 Political donations	2	0.2%	135 School grades (class levels)	1	0.1%
36 Sexual orientations	9	1.1%	86 Political views	2	0.2%	136 Spouses names	1	0.1%
37 Avatars	8	0.9%	87 Professional skills	2	0.2%	137 Tattoo status	1	0.1%
38 Email messages	8	0.9%	88 Support tickets	2	0.2%	138 Taxation records	1	0.1%
39 Family structure	8	0.9%	89 Survey results	2	0.2%	139 Travel habits	1	0.1%
40 Instant messenger identities	8	0.9%	90 Telecommunications carrier	2	0.2%	140 Travel plans	1	0.1%
41 Bank account numbers	7	0.8%	91 Address book contacts	1	0.1%	141 User statuses	1	0.1%
42 Credit cards	7	0.8%	92 Appointments	1	0.1%	142 Utility bills	1	0.1%
43 Drinking habits	7	0.8%	93 Apps installed on devices	1	0.1%	143 Vehicle identification numbers (VINs)	1	0.1%
44 Occupations	7	0.8%	94 Astrological signs	1	0.1%	144 Warranty claims	1	0.1%
45 Passport numbers	7	0.8%	95 Audio recordings	1	0.1%	145 Work habits	1	0.1%
46 Smoking habits	7	0.8%	96 Beauty ratings	1	0.1%	146 Years of professional experience	1	0.1%
47 Social security numbers	7	0.8%	97 Biometric data	1	0.1%			
48 Time zones	7	0.8%	98 Buying preferences	1	0.1%			
49 Job applications	6	0.7%	99 Car ownership statuses	1	0.1%			
50 Relationship statuses	6	0.7%	100 Career levels	1	0.1%			

Note: This table breaks down the 146 data types compromised across the 857 breaches, listing the number (#) and percentage (%) of breaches involving each data type. A breach can expose multiple data types.

Table SI3. Classification of the 38 Serious Data Classes in Breaches

#	Category	#	Category
1	Audio recordings	20	MAC addresses
2	Auth tokens	21	Mothers' maiden names
3	Bank account numbers	22	Nationalities
4	Biometric data	23	Partial credit card data
5	Browsing histories	24	Partial dates of birth
6	Chat logs	25	Passport numbers
7	Credit card CVV	26	Password hints
8	Credit cards	27	Passwords
9	Credit status information	28	Personal health data
10	Drinking habits	29	Photos
11	Driver's licenses	30	PINs
12	Drug habits	31	Places of birth
13	Email messages	32	Private messages
14	Encrypted keys	33	Security questions and answers
15	Government issued IDs	34	Sexual fetishes
16	Health insurance information	35	Sexual orientations
17	Historical passwords	36	SMS messages
18	HIV statuses	37	Social security numbers
19	Login histories	38	Taxation records

Note: List of data classes that, if exposed or leaked in data breaches, will pose serious security risks to the individual (e.g., used to gain unauthorized access to other accounts, commit identity theft, or personally sensitive information). See also Table 2.

Table SI4. Country-level predictors of estimated country fixed effects

	Dependent variable: estimated country fixed effect			
	Any breach		Serious breach	
	(1)	(2)	(3)	(4)
Log GDP per capita	0.102*** (0.021) [0.060, 0.140]	0.001 (0.044) [-0.085, 0.097]	0.055** (0.016) [0.024, 0.094]	-0.017 (0.028) [-0.077, 0.045]
Digitization (EGDI)		0.457 (0.352) [-0.248, 1.136]		0.516+ (0.287) [-0.031, 1.120]
English official language		0.213*** (0.049) [0.121, 0.312]		0.166** (0.057) [0.060, 0.282]
Global Cybersecurity Index (GCI)		0.001 (0.001) [-0.002, 0.003]		-0.001 (0.001) [-0.003, 0.000]
Electoral democracy (V-Dem)		0.436*** (0.121) [0.133, 0.693]		0.246** (0.087) [-0.006, 0.448]
Constant	-0.561** (0.186) [-1.159, -0.306]	-0.337 (0.215) [-1.035, -0.050]	-0.237+ (0.138) [-0.728, -0.100]	-0.054 (0.133) [-0.545, 0.098]
Dependent variable mean	0.362	0.362	0.261	0.261
Countries	52	52	52	52
R ²	0.358	0.610	0.244	0.501
Adjusted R ²	0.345	0.567	0.229	0.447

Note: OLS point estimates using current, unrounded first-stage country fixed effects. Parentheses contain HC3 standard errors. Brackets contain percentile 95% intervals from 999 hierarchical bootstrap draws that resample countries and then email addresses within countries and re-estimate both stages. English official language is an indicator; GCI is the 0–100 Global Cybersecurity Index of national commitments; EGDI and V-Dem range from 0 to 1. Stars use HC3 p-values. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table SI5. Estimated country fixed effects (ranked by serious breaches)

#	ISO	Country	Serious breaches	Breaches
1	AUS	Australia	0.891	0.941
2	GBR	UK	0.798	0.83
3	JEY	Jersey	0.788	0.878
4	HKG	Hong-Kong	0.633	0.659
5	DNK	Denmark	0.458	0.687
6	FIN	Finland	0.427	0.695
7	NOR	Norway	0.421	0.723
8	NLD	Netherlands	0.406	0.736
9	NZL	New-Zealand	0.391	0.599
10	ZAF	South-Africa	0.384	0.518
11	IND	India	0.357	0.653
12	URY	Uruguay	0.339	0.355
13	SGP	Singapore	0.334	0.499
14	LUX	Luxembourg	0.333	0.407
15	NIC	Nicaragua	0.33	0.313
16	PYF	French-Polynesia	0.33	0.321
17	KEN	Kenya	0.302	0.581
18	SYC	Seychelles	0.294	0.284
19	UGA	Uganda	0.293	0.316
20	CAN	Canada	0.293	0.865
21	MLT	Malta	0.291	0.612
22	BRA	Brazil	0.29	0.717
23	ARG	Argentina	0.28	0.484
24	BEL	Belgium	0.278	0.43
25	NGA	Nigeria	0.264	0.377
26	BTN	Bhutan	0.263	0.254
27	GGY	Guernsey	0.261	0.273
28	GRC	Greece	0.26	0.491
29	ZWE	Zimbabwe	0.243	0.212
30	EST	Estonia	0.243	0.323
31	HUN	Hungary	0.241	0.319
32	ITA	Italy	0.231	0.354
33	CYP	Cyprus	0.208	0.502
34	BIH	Bosnia-and-Herzegovina	0.202	0.217
35	MDA	Moldova	0.2	0.213
36	SUR	Suriname	0.198	0.188
37	COL	Colombia	0.196	0.192
38	AND	Andorra	0.193	0.182
39	BMU	Bermuda	0.185	0.206
40	NAM	Namibia	0.183	0.174
41	GRL	Greenland	0.181	0.171
42	GTM	Guatemala	0.169	0.166
43	GEO	Georgia	0.168	0.255
44	BGR	Bulgaria	0.163	0.207
45	ARM	Armenia	0.163	0.296
46	LKA	Sri-Lanka	0.154	0.145
47	CMR	Cameroon	0.154	0.145
48	MKD	Macedonia	0.154	0.145
49	RWA	Rwanda	0.154	0.145
50	ZMB	Zambia	0.154	0.16
51	ROU	Romania	0.148	0.159
52	BLR	Belarus	0.145	0.135
53	TZA	Tanzania	0.134	0.126
54	SVK	Slovakia	0.13	0.19
55	NPL	Nepal	0.129	0.132
56	ALB	Albania	0.127	0.12
57	PNG	Papua-New-Guinea	0.107	0.115
58	KOR	South-Korea	0.074	0.084
59	IRN	Iran	0.025	0.045

Note: Estimated country fixed effects, partialing out personal/official email classification and decade fixed effects. The serious-breach column comes from Model (3) of Table 6; the any-breach column comes from Model (1). Values are rounded only for display; downstream analysis uses the unrounded RDS artifact.

Breach provenance

The headline breach rate pools incidents that differ in what they imply about the account holder. We therefore assign every breach in the catalogue to exactly one of three classes according to how the data was exposed. *Broker / scrape aggregation* covers data brokers, scrapers and enrichment services: the address was held by a third party the politician had no relationship with. *Credential combolist* covers aggregated credential dumps, spam corpora and stealer logs, whose constituent sources are mixed or unknown. *Service compromise* covers breaches of a service the account holder plausibly held an account with, and is the only class that supports a statement about a politician’s own accounts.

Assignment is rule-based on HIBP’s `IsSpamList` flag and the text of the breach description. Four of the 857 breaches are assigned by hand where the rules misclassify them; the overrides and their written rationales ship in the replication package.

[Table SI6](#) reports the result. Of the 33.0% of addresses that appear in at least one breach, 15.9% appear in a service compromise, 23.5% in broker or scrape aggregation, and 15.7% in a credential combolist. These shares do not sum to 33.0% because an address can appear in more than one class.

The implication is that the headline rate substantially overstates how often politicians’ own accounts have been compromised. More addresses are exposed because a commercial aggregator held them than because a service the politician used was breached, and an address can enter a broker database through no action of the politician at all. This also helps explain the official/personal gradient reported in the main text: official addresses are published by design, which is precisely what makes them attractive to harvest.

Table SI6. Breach appearances by how the data was exposed

Provenance	Breach appearances	Addresses	% of appearances	% of addresses
Broker / scrape aggregation	5,570	2,915	44.5%	23.5%
Credential combolist	3,279	1,949	26.2%	15.7%
Service compromise	3,672	1,969	29.3%	15.9%

Note: Every breach in the catalogue is assigned to exactly one class: *broker / scrape aggregation* (a data broker, scraper or enrichment service held the address, and the account holder had no relationship with it), *credential combolist* (aggregated credential dumps, spam corpora and stealer logs of mixed or unknown provenance), and *service compromise* (a service the account holder plausibly held an account with was itself breached). Assignment is rule-based on HIBP’s `IsSpamList` flag and the breach description; four breaches are assigned by hand and the overrides, with written rationales, ship in the replication package as `data/breach_taxonomy_overrides.csv`. Column shares of addresses do not sum to the overall breached share because one address can appear in more than one class. HIBP indexes only breaches that have become public and been ingested, and we observe at most one address per politician for most of the sample, so every figure is a lower bound.

The four hand-coded assignments are the only discretionary element of this classification, and [Section 4.4](#) reports how much they move the result.

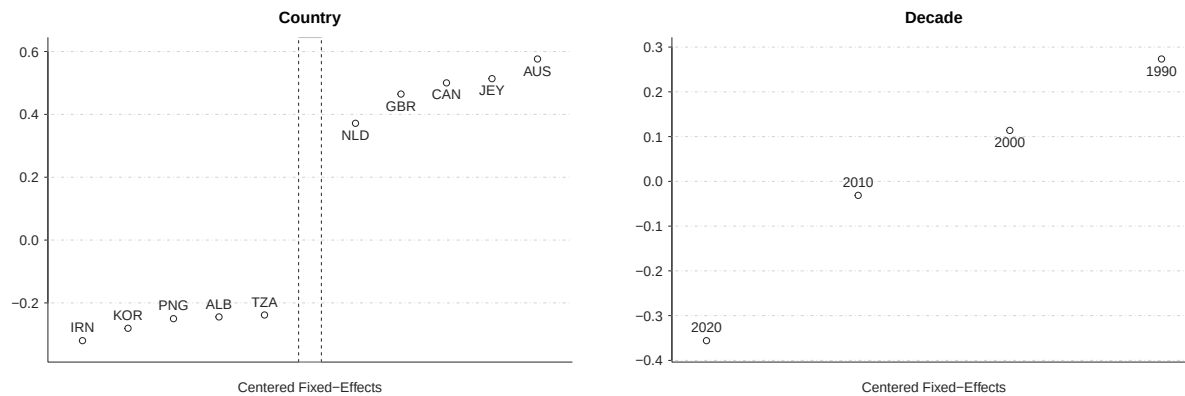


Figure SI2. Country and decade mean-centered fixed effects of the probability of a breach. This figure reports the top five and bottom five estimated fixed effects for the country (left panel) and the fixed effects for a decade (right panel) from Model (1) in Table 6. All plotted effects are centered such that the mean is zero. See Figure 1 for the same plot for serious breaches. See Table SI5 for all countries' estimated effects.

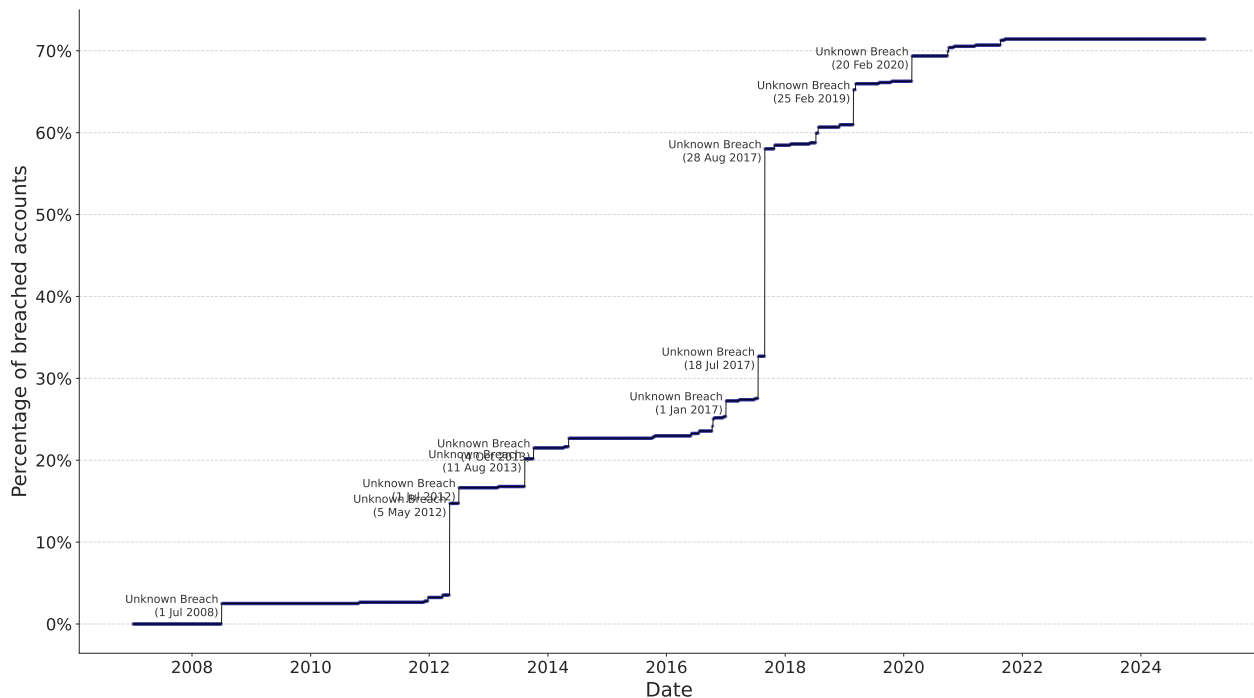


Figure SI3. Breach rate of a fixed cohort of politician emails over time. Similar to Figure 3, except that this plot follows a fixed cohort of $n = 679$ politician emails where the earliest known date is before 2007 (the first recorded HIBP breach is on 12 July 2007, Table SI1). The top panel annotates a few of the largest jumps in percentage and the corresponding breach incidents.

Comparison with population samples 5,000 US adults whose addresses were queried against HIBP in 2018. Three harmonisations are required before the samples are comparable.

To place the level of exposure in context we compare against Sood and Cor (2019), a survey of

First, the catalogue. The two studies queried HIBP at different dates, against catalogues of 293 and 857 breaches respectively, and the smaller is a strict subset of the larger. We restrict both samples to the 293 breaches they share.

Second, the exposure window. The common catalogue spans 2007–2018, so restricting to it also bounds the period over which exposure could have occurred. Because politician addresses are comparatively recent, we further restrict them to addresses whose earliest observed legislative term began in 2018 or before, leaving 8,488 addresses.

Third, address type. Most politician addresses in our sample are institutional, whereas survey respondents supplied personal addresses. We therefore also report politicians restricted to personal-domain addresses.

[Table SI7](#) reports all three. Under the fullest set of restrictions, 17.6% of politician addresses and 24.6% of politician personal addresses appear in at least one of the common breaches, against 82.8% of US adults.

We stress that this is a benchmark and not a control. US addresses are over-represented in HIBP’s corpora, and our sample contains no US legislators, so the difference in levels is confounded by coverage and is not an estimate of relative risk.

Subject to that caveat, the comparison does not support the view that politicians are unusually exposed. The direction is the opposite, and it survives the two adjustments available to us: matching the exposure window raises the politician rate only from 15.3% to 17.6%, and restricting to personal addresses raises it to 24.6%, still well below the benchmark. The composition differs as sharply as the level: 71.7% of US adults appear in a service compromise against 7.7% of politicians, so the general-population sample is exposed largely through accounts its members held, and ours largely through addresses others published. We draw no conclusion about relative risk from this. We do conclude that a prevalence figure of this kind carries no evidential weight until it is compared against something, and that reporting one unanchored invites the reader to supply a comparison that the data does not support.

Stealer-log exposure

HIBP loads a small number of infostealer and malware corpora into the ordinary breach dataset. Several originate in documented law-enforcement operations: the *Emotet* data was supplied following the FBI and Dutch NHTCU takedown, and the *Qakbot* data following the US Department of Justice operation. These differ from the

Table SI7. Politicians and US adults on a common breach catalogue and a common exposure window

Sample	<i>n</i>	Any	Serious	Service	Broker	Combolist	Stealer log
Politicians	8,488	17.6%	17.2%	7.7%	0.9%	14.6%	1.0%
Politicians (personal addr.)	1,617	24.6%	23.9%	17.1%	0.7%	17.1%	2.0%
US adults (YouGov)	5,000	82.8%	68.1%	71.7%	2.7%	62.9%	7.0%

Note: All samples are restricted to the 293 breaches common to this study and Sood and Cor (2019), whose catalogues are nested. That catalogue spans 2007–2018, so the restriction also bounds the exposure window; politicians are therefore further restricted to addresses whose earliest legislative term began in 2018 or before. The final row restricts politicians to personal-domain addresses, the closest match to the personal addresses YouGov respondents supplied. **This is a benchmark, not a control.** US addresses are over-represented in HIBP corpora and the politician sample contains no US legislators, so the difference in levels is not identified and should not be read as an estimate of relative risk. HIBP indexes only breaches that have become public and been ingested, and we observe at most one address per politician for most of the sample, so every figure is a lower bound.

rest of the catalogue in that appearance indicates credentials captured from an infected machine rather than a third party holding the address.

Table SI8 reports exposure across the nine such corpora. 287 addresses, or 2.32% of the sample, appear in at least one. Of those, 216 appear in one corpus, 40 in two, 25 in three and 6 in four. The rate is 5.54% among personal addresses and 1.28% among official addresses.

These figures are lower bounds even by the standards of the rest of this paper. HIBP maintains a separate stealer-log dataset, larger than the portion loaded as breaches, which is queryable only by parties who have verified control of the address or of its email domain. We can therefore observe whether an address appears in the corpora loaded as breaches, but not the websites against which the credentials were

captured. The corpora we do observe also differ in provenance: those supplied after law-enforcement operations are curated victim lists, whereas those collated from criminal distribution channels are bulk dumps, and it is the latter that account for nearly all of the appearances we record. An infection whose logs were never dumped publicly is invisible to us.

This channel is small but is the one that speaks most directly to the security of politicians themselves, because appearance indicates credential capture from a compromised device rather than a third party holding the address. Two features are worth noting. Repeat appearance is common: 71 of the 287 exposed addresses appear in more than one corpus, which is difficult to explain as incidental inclusion. And the rate is roughly four times higher for personal than for official addresses, though this may reflect which address is entered into a browser on a personal device rather than differential infection of politicians’ machines. We also note that credentials captured this way typically include session cookies, which authenticate without invoking a second factor; this is the specific threat that motivates our recommendation of phishing-resistant multi-factor authentication rather than one-time codes.

Table SI8. Politician addresses appearing in stealer-log corpora

Corpus	Breach date	Addresses	% of sample	95% CI
TelegramCombolists	2024-05	164	1.32%	[1.14, 1.54]
TrikSpamBotnet	2018-06	95	0.77%	[0.63, 0.94]
NazApi	2023-09	63	0.51%	[0.40, 0.65]
StealerLogsJan2025	2025-01	40	0.32%	[0.24, 0.44]
TelegramStealerLogs	2024-07	31	0.25%	[0.18, 0.36]
RedLineStealer	2021-12	2	0.02%	[0.00, 0.06]
Emotet	2021-01	0	0.00%	[0.00, 0.03]
NamelessMalware	2020-01	0	0.00%	[0.00, 0.03]
Qakbot	2023-08	0	0.00%	[0.00, 0.03]
<i>Any stealer corpus</i>		287	2.32%	[2.07, 2.60]
personal addresses		167	5.54%	[4.78, 6.42]
official addresses		120	1.28%	[1.07, 1.53]

Note: HIBP loads some infostealer and malware corpora into the ordinary breach dataset; several originate in documented law-enforcement seizures (Emotet, from the FBI and Dutch NHTCU takedown; Qakbot, from the US Department of Justice operation). Unlike the rest of the catalogue, appearing in one indicates credentials captured from an infected machine rather than a third party holding the address. Intervals are Wilson binomial intervals. Per-corpus counts sum to more than the total because addresses appear in more than one corpus. **These are lower bounds even by the standards of the rest of this paper:** HIBP’s dedicated stealer-log endpoint indexes more data, and it cannot be queried for this sample at any subscription tier, because access requires verified control of the address or of its email domain.

Sensitivity to construction choices

Two steps in building these figures are judgement rather than arithmetic. We report what each costs rather than asserting that either is innocuous.

The first is the classification of four breaches by hand, where the rules described above misassign them; those four carry roughly 2.7% of all appearances. [Table SI9](#) reverts each in turn to the class the rules would have given, and then all four together. No share moves by more than 1.07 percentage points under any specification, and the movement is concentrated almost entirely in one breach (*DemandScience*). Broker aggregation exceeds service compromise in ev-

ery specification, including the one that reverts all four hand-coded rows; the replication package fails the build if that ordering ever ceases to hold.

The second is the rule for collapsing addresses observed in both sources, which affects 531 of the 12,385 addresses. [Table SI10](#) varies it. Taking the maximum, the sum, or the first of an address’s breach counts gives identical results, because the duplicate rows are structured so that one carries every breach and the other carries none; only taking the minimum differs, and it selects the row that observed no breach data. The choice of legislative start year changes how many addresses fall inside a given exposure window, but leaves every rate unchanged. The seriousness threshold, the third such choice, is reported separately in [Table 3](#).

Table SI9. Provenance shares under alternative treatment of the hand-coded breaches

Specification	Appearances	Broker	Combolist	Service
<i>As published</i>		23.54%	15.74%	15.90%
revert B2BUSABusinesses	76	23.54%	15.79%	15.90%
revert DemandScience	205	23.04%	15.74%	16.97%
revert Epik	35	23.58%	15.74%	15.77%
revert LuminPDF	49	23.76%	15.74%	15.70%
<i>All four reverted</i>		23.32%	15.79%	16.66%

Note: Four of the 857 breaches are assigned to a provenance class by hand where the rules misclassify them; they carry roughly 2.7% of all appearances. Each row reverts the named breach to the class the rules would have assigned, holding the other three at their coded values; the final row reverts all four. Broker aggregation exceeds service compromise in every specification, and the build fails if that ceases to hold.

Email classification patterns

Table SI10. Headline rates under alternative deduplication rules

Year / count rule	Breached	Serious	Mean breaches	Addresses by 2018
min / max (<i>used</i>)	33.05%	21.58%	1.011	8,488
min / sum	33.05%	21.58%	1.011	8,488
min / first	33.05%	21.58%	1.011	8,488
min / min	30.30%	20.36%	0.949	8,488
max / max	33.05%	21.58%	1.011	7,957
max / sum	33.05%	21.58%	1.011	7,957
max / first	33.05%	21.58%	1.011	7,957
max / min	30.30%	20.36%	0.949	7,957

Note: The address-level file stores one row per source, so 531 of the 12,385 addresses appear twice and must be collapsed. Rows vary the rule used for the legislative start year (minimum or maximum across an address's rows) and for the breach counts. Taking the maximum, the sum, or the first value gives identical results, because the duplicate rows are structured so that one carries every breach and the other carries none; only the minimum differs, and it selects the row that observed no breach data. The start-year rule changes how many addresses fall inside a given exposure window but does not change any rate.

Table SI11. Patterns for Core Government Domains

Regex Pattern	Notes
\.gov\$	English pattern
\.gov\[a-z]{2,3}\$	e.g., gov.uk, gov.sg
\.gob\$	Spanish pattern
\.gouv\$	French pattern
\.go\[a-z]{2,3}\$	East Asian pattern
\.gc\[a-z]{2,3}\$	Canada
\.fed\[a-z]{2,3}\$	Federal (e.g., fed.us)
\.mil\$	Military
\.admin\$	Administrative (e.g., admin.ch)
\.bund\$	German federal
\.fgov\$	Belgian federal
\.nic\.	India (e.g., sansad.nic.in)
\.mil\.za\$	Dept. of Defence of South Africa

Table SI12. Patterns for Political Parties and Organizations

Regex Pattern	Notes
dphk\.org\$	Democratic Party (HK)
bjpanda\.org\$	BJP and allies (India)
iy\.in\$	Indian Youth Congress
da-mp\.org\.za\$	Democratic Alliance (S. Africa)
ifp\.org\.za\$	Inkatha Freedom Party (S. Africa)
fondazione craxi\.org\$	Craxi Foundation (Italy)
partitodemocratico\.it\$	Democratic Party (Italy)
ecolo\.be\$	Ecolo (Belgium)
liberal\.org\.hk\$	Liberal Party (HK)
dab\.org\.hk\$	Democratic Alliance (HK)
pap\.org\.sg\$	People's Action Party (Singapore)
prpg-grc\.sg\$	Grassroots organizations (Singapore)
wp\.sg\$	Workers' Party (Singapore)
udm\.org\.za\$	United Democratic Movement (S. Africa)

Table SI13. Patterns for Commercial or Personal Domains

Regex Pattern	Notes
\.com\.	Commercial
\.net\.	Network
\.co\.	Company
\.biz\.	Business
\.info\.	Informational
\.corp	Corporate domain
libero\.it\$	Italian commercial email provider
\.ltd	Limited company domain
\.inc	Incorporated company domain
\.enterprise	Enterprise domain
\.business	Business domain
yahoo	Common email provider
hotmail	Common email provider
gmail	Common email provider

Table SI14. Patterns for Known Parliaments and Ministries

Regex Pattern	Notes
ft\.\$	Danish Parliament
senato\.\$	Italian Senate
camera\.\$	Italy
stortinget\.\$	Norwegian Parliament
tweedekamer\.\$	Netherlands
cdep\.\$	Romania
eduskunta\.\$	Finland
assnat\.\$	Cameroon
riigikogu\.\$	Estonia
sobranie\.\$	Macedonia
dekamer\.\$	Belgium
chd\.\$	Luxembourg
lachambre\.\$	Belgium
dna\.\$	Suriname
inatsisartut\.\$	Greenland
nanog\.\$	Greenland government
um\.\$	Danish Ministry of Foreign Affairs
fm\.\$	Danish Ministry of Finance
skm\.\$	Danish Prime Minister's Office
sum\.\$	Danish Ministry of Health
trm\.\$	Danish Ministry of Transport
uim\.\$	Danish Immigration Ministry
jm\.\$	Danish Ministry of Justice
kum\.\$	Danish Ministry of Culture
bm\.\$	Danish Ministry of Employment
uvm\.\$	Danish Ministry of Education
stm\.\$	Danish Prime Minister's Office
aeldremin\.\$	Danish Ministry for Elderly
fvm\.\$	Danish Ministry of Food
evm\.\$	Danish Ministry of Energy
efkm\.\$	Danish Environment Ministry
km\.\$	Danish Church Ministry
em\.\$	Danish Business Ministry
oim\.\$	Other Danish Ministry
sm\.\$	Social Ministry Denmark
ufm\.\$	Higher Education and Science
mfvm\.\$	Food, Agriculture and Fisheries
mssb\.\$	Danish Ministry for Gender Equality
mfa\.\$	Greek Ministry of Foreign Affairs
istruzione\.\$	Italian Ministry of Education
nic\.\$	Indian Government (generic)
nrsr\.\$	Slovak National Council
nationalassembly\.\$	Seychelles
parliran\.\$	Iranian Parliament
majles\.\$	Iranian Parliament
europarl\.\$	European Parliament
giunta\.\$	City of Verona
comune\.\$	City of Pescara
comune\.\$	City of Catania
regione\.\$	Liguria Region
regione\.\$	Lazio Region
provincia\.\$	Venice Province
andorra\.\$	Andorra
@.*\.\$	Catch-all for .gov.*
@.*parliament\.\$	Parliament domains
@.*senat\.\$	Senate domains
@.*assembly\.\$	Assembly domains
@.*congress\.\$	Congress domains
@.*ministry\.\$	Ministry domains
@.*cabinet\.\$	Cabinet domains
@.*bureau\.\$	Bureau domains
\bparliament\b	Contains "parliament"
\bparlament\b	Contains "parlament"
\bparlamento\b	Contains "parlamento"
\bparl\b	Contains "parl"
\bsenat\b	Contains "senat"
\bsenado\b	Spanish Senate
\bassembly\b	Assembly variant
\bassemblee\b	French Assembly
\basamblea\b	Spanish Assembly
\bcongress\b	Congress variant
\bcongreso\b	Spanish Congress
\bministry\b	Ministry keyword
\bcabinet\b	Cabinet keyword
\bbureau\b	Bureau keyword
\bgovernment\b	Government keyword